

Rama krishna Boorgadda (RK)

919-532-6376 | boorgaddar@gmail.com

Professional Summary

AI Engineering leader with over 19 years of experience and more than 8 years focused on generative AI, LLMs, and multi-agent systems. Proven track record of architecting and deploying enterprise-grade RAG and agentic AI applications using Python, LangChain, vector databases, and cloud services. Skilled at defining technical roadmaps, guiding cross-functional teams, and building scalable AI pipelines to solve complex business challenges.

Academic Profile

Holy Jesus & Mary PG College, Osmania University
MCA

2004 - 2006

SSR College, Osmania University
BSc computers

2000 - 2002

Technical Skills

- **Generative AI:** LLMs, Diffusion Models, GANs, Prompt Engineering, Multimodal AI, Retrieval-augmented Generation (RAG), Multi-agent Systems
- **Java Skills:** Java/J2EE up to date
- **Application frameworks:** Spring Boot, Spring Security, Spring Enterprise Integrations, Spring Data Flows, Struts, Hadoop etc
- **Technical Stack:** Python, PyTorch, TensorFlow, Hugging Face, Lang Chain, Kubernetes, Langgraph
- **Cloud & Scaling:** AWS, GCP, Azure, Large-scale Data Pipelines, MLOps, AWS SageMaker, AWS Bedrock
- **Middleware Technologies:** OSB, ALDSP (ODSI), BPEL, Oracle Event Processing
- **AI/ML:** Python, TensorFlow, Keras, Pandas, Machine Learning
- **Vector Databases:** Chroma, Pinecone, Mongo
- **Data Base stack:** Oracle, MySQL, Postgres, DB2, Mongo, SQL Server
- **Design Patterns:** J2EE Patterns, SOA Patterns, Enterprise Integration patterns, GenAI Application Architecture
- **Web Technologies:** Angular, ReactJS
- **Programming Languages:** C, C++, .Net
- **ETL:** Kettle Open source ETL tools
- **Messaging:** IBM WebSphere, Rabbit MQ, Kafka
- **Cloud:** Docker, Kubernetes, AWS
- **Gateways:** Layer 7, IBM Data Power and Apigee
- **AI Tools:** Claude Code, Codex, Cursor, Windsurf
- **AI Observability & Evaluation Tools:** Langsmith, Langfuse

Work Experience

FreedieMac

Jun 2024 - Present

Director/Sr Principal AI Engineer

Developed a full-stack, enterprise-grade AI chat application to serve as an intelligent assistant with a seamless, context-aware “Intellisense” experience. The application supports multi-turn conversations, file uploads, and real-time code/text suggestions to significantly enhance user productivity. The core of the application is a powerful Agentic Retrieval-Augmented Generation (RAG) system that provides highly accurate and relevant responses.

- Designed and implemented a robust GenAI application architecture for a full-stack, enterprise-grade AI chat application using Python for the backend and Streamlit for the interactive, real-time frontend.
- Developing the overall AI and LLM adoption strategy, setting goals, and aligning them with business objectives.
- Overseeing the development and refinement of NLP/LLM models, including RAG (Retrieval-Augmented Generation), fine-tuning, and prompt engineering.
- Engineered an advanced multi-agent RAG pipeline by integrating a Hugging Face LLM as the core reasoning engine and leveraging orchestration frameworks like Lang Chain and smolagents.
- Developed and managed a high-performance vector database using Pinecone/Qdrant for efficient similarity search and retrieval of contextual information to support the RAG system.
- Built a responsive and intuitive user interface with Streamlit, implementing features like streaming responses, context-aware autocomplete, and seamless file uploads to create a dynamic and engaging user experience.
- RAG evaluation metrics implemented to measure accuracy.
- Implemented LORA and QLORA Fine tuning techniques.
- Integrated the application with a wide range of external tools and data sources by implementing the Model Context Protocol (MCP), enabling the AI agent to perform complex, real-world actions.
- Implemented a robust and secure authentication system to support multiple users in an enterprise environment, ensuring data privacy and personalized user sessions.

- Designed a scalable and resilient system architecture to handle a growing user base and expanding knowledge base.
- Created and configured CI/CD pipelines to automate the build, testing, and deployment processes, ensuring continuous and reliable delivery of new features.
- Sentiment Analysis for loan buyers and sellers using NLP Techniques.
- Utilized Langsmith and Langfuse for evaluation and optimization of LLM outputs, employing LORA and QLORA techniques.

FreddieMac

Nov 2023 - Jun 2024

Senior AI Engineer

Developed a Retrieval-Augmented Generation (RAG) based intelligent Q&A system to provide instant, accurate answers from the complex Freddie Mac Seller/Service Guide. The application empowers mortgage professionals, including lenders, servicers, and underwriters to quickly find specific regulatory requirements, reducing research time and improving compliance. The system uses OpenAI's embedding and completion models to perform semantic search on a pre-processed knowledge base of the guide's content, delivering context-aware responses through a simple API.

- Developed and maintained a Python-based Gradio Interface to serve the Q&A system, handling user queries and returning AI-generated answers.
- Engineered the core Retrieval-Augmented Generation (RAG) pipeline, which involved loading a pre-computed knowledge base of 225+ FAQs and their corresponding vector embeddings from the Freddie Mac Seller/Service Guide.
- Built a Streamlit app for external users to interact with the Q&A system.
- Leveraged the OpenAI API for key AI functionalities: used the text-embedding-ada-002 model to generate vector embeddings for user queries and the text-davinci-003 model for generating context-aware answers.
- Implemented a high-performance semantic search capability using NumPy to calculate cosine similarity (via dot product) between user query embeddings and the pre-calculated document embeddings, ensuring fast and accurate retrieval of relevant context.
- Designed and implemented a dynamic prompt engineering strategy that constructs context-rich prompts for the completion model, including retrieved document sections and instructions to ensure truthful, in-context responses.
- Managed API costs and performance by implementing token-aware context construction, using Tiktoken and Transformers tokenizers to ensure prompts stayed within the model's token limits.
- Processed and structured the raw content from the Freddie Mac guide into a clean dataset using Pandas, creating a knowledge base of questions, answers, and context for the RAG system.
- Containerized the Flask application using Docker for consistent deployment and scalability.
- Inspected prompts and responses to block malicious injections, jailbreaks, and sensitive data leakage using Amazon Bedrock guardrails and AI tools codex, claude code, cursor, and windsurf.

Wells Fargo

Oct 2021 - Nov 2023

Senior AI Engineer

Designed, architected, and delivered multiple Regression and classification Problem statements and Generative AI applications, including:

- Designed and implemented an efficient multi-class Credit risk assessment and Loan approval.
- Generative AI application designed for the log analysis and label the group using unsupervised machine learning technique like DBSCAN.
- Designed, built and implemented regression models like Decision Tree, Random Forest and linear regression for credit card charge predictions.
- AI-driven financial advisory assistant combining client data, real-time market feeds, and internal research to provide personalized investment insights

Schneider

Jan 2020 - Oct 2021

Solution Architect

Standard flows have been incorporated and designed for the Telematics Redesign for Schneider. Flows handled for the Schneider.

- Vehicle Faults: used unsupervised machine learning techniques to group the users regarding notification text.
- Critical Events: find the drivers and group them based on major events; notification emails get sent from customer care.
- Improved response time turnaround from customer care to drivers by implementing the above techniques.
- Implemented DBSCAN clustering techniques.
- Driver behavior analysis and safety scoring classification problem statement. Machine learning algorithms classify safe vs. risky drivers.
- Classification Report to measure accuracy, precision, recall, and F1

Arrow Partners

Apr 2017 - Jan 2020

Architect

All the Standard b2B flows from EBS to B2B partners and from partners to inbound B2B. Worked on providing the solution on Cloud Integrations with Oracle Cloud EBS. End-to-end Implementation for Oracle Cloud EBS with B2B Partners with OIC and ICS. Flows handled for partner Specific Integrations: CustPO, SupplierPO, ConsumptionAdvice, SupplierRemittance, CustPOChng, DPP Foundation Pack, Rest Implementation to partners on HTTP or JSON, SupplierPoChng, Setting up SFTP from OIC to integrate, Scheduled Integration Flows in OIC, Expertise with Oracle ICS and integrating Oracle and non-Oracle platforms.

- Designed OAGIS and leveraged standards in SOA Integrations.
- Proposed Reprocessing Milestones.

- Oracle ICS and integrating Oracle and non-Oracle platforms.
- Worked on Solutioning the OSB for Partner specific Integrations with REST and JSON.
- Created SSH standards for authenticating & authorizing the partners.
- Architected upgrades from 11g to 12C.
- Excellent knowledge of Operation Production Support with large mission critical systems including WebLogic, SOA, Apache, and UNIX.
- Applied patches to Oracle SOA Suite 12c Environments, involved DB sizing for 12c upgrade.
- Installed the Foundation Pack for DPP.
- Prepared plan for Datacenter moves. Involved in DR.
- Good experience in Configuration, Performance Tuning, and Troubleshooting of SOA Environment Service Engines, Adapters, BAM.
- Lead upgrade for NA, Asia Pack & Europe successfully.

KPN **Jun 2015 - Mar 2017**
Bangalore
Consultant/Architect

Cloud Store is a new online web shop for SWOL services, using Parallels for provisioning products and services. Products can be added to Parallels using pre-built provision plugins. Previously bought products and services can be managed using integrated self-care providing a view on Parallels.

- Worked on upgrade from 11g to 12c SOA.
- Replacing legacy systems to work with WSDLs for better orchestration.
- Handled entire Upgrade and decomposed work needed in SOA.
- Filled gaps for backward compatibility during upgrade.
- Ensured all services authenticated with WS-Security standard.
- Worked On separate error handling framework.
- Defined upgrade plan from 11g to 12C.
- Analyzed open items and dependencies, maintained backward compatibility.
- Designed and implemented architectural proof for integrating Cloudstore with Parallel system.
- Replaced Java Embedding with WSDL for loosely coupling.
- Used MFT for binary file transfer from Cloud store to Parallel.
- GetProductAndPriceListCloudStoreReqABCSImpl
- InternalGetProductAndPriceListCloudStoreReqABCSImpl
- ProductAndPriceListEBS
- ProductAndPriceListEBF
- GetSalesCategoryListParallelsProvABCSImpl
- GetPlanListAvailable4SalesCategoryParallelsProvABCSImpl
- GetPlanPeriodListParallelsProvABCSImpl
- GetPlanResourceListParallelsProvABCSImpl
- GetEnablingResourceByPlanListProvABCSImpl
- GetPlanParamsGetAndValidateParallelsProvABCSImpl
- Mentored juniors and SMEs.
- Successfully delivered upgrade with improved performance

General Electric (USA) **Jun 2012 - Jan 2015**
Technical Architect/Manager

GE Appliances ERP Engagement involved consolidating different legacy apps and custom ERP solutions into one standard Oracle ERP and supporting apps. Primary stack comprised: Supply Chain Management (SCM): Oracle and Proficy (from GE Intelligent Platforms); Order Ship Bill: Oracle; Financial Resource Management (FRM): Oracle; Product Lifecycle Management (PLM): Windchill; Customer Relationship Management (CRM): Salesforce and Siebel (Oracle product).

- Technical Architect/Manager responsible for managing and solutioning integration projects in Oracle SOA and EDI.
- Lead/developed technical solution design.
- Meeting project commitments and communications with sponsors, stakeholders.
- Coordinated with client and offshore team ensuring timely deliverables.
- Understanding business requirements and contributing in high-level design.
- Client coordination for need identification.
- Facilitated defining project goals and resource requirements, conflict resolution.
- Ensured products and data designed/integrated cohesively with proper documentation.
- Worked as business partner with ERP leaders and users to improve business processes.
- Drove thorough test cases, unit testing, and integrated testing.

Oracle Corporation **Jan 2009 - Jun 2012**
Technical Lead

Oracle Pedigree and Serialization Manager (OPSM) manages globally unique serials and serial transactions for serialized item verification. Requires integration with systems managing Manufacturing, Packaging, Shipping, Receiving, and Returns. Initial integration records/tracks: SyncItemLotList, AdvanceShipmentNoticeList, ShipmentReceiptList.

- Requirements gathering.
- Analysis and Design of application.
- Prepared TDD.
- Worked on development of BPEL, SOA, Mediator.

BROCADE

Apr 2007 - Jan 2009

Developer

Integration ensures location information created/updated in E-Business suite transfers to OTM. SyncLocationList: Raised business event handled by Oracle Application Adapter, data enrichments via BPEL, routed through EBS.

- Requirements gathering.
- Analysis and Design of application.
- Prepared TDD.
- Worked on development of BPEL.